

Electronic Supplementary Information

Native implementation of the machine-learned Skala exchange–correlation functional in CP2K:
Unified one-centre reconstruction for molecular and condensed-phase calculations

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S1 Scope and Notation

This Electronic Supplementary Information (ESI) documents the numerical definition and validation of the molecular and condensed-phase comparisons. Self-consistent-field (SCF) convergence, basis resolution, and grid resolution are assessed separately. The SCF solvers are orbital transformation (OT) and diagonalization. Electronic occupations and the overlap-matrix trace define the explicitly treated particle number. Integration on the model quadrature provides an additional discretization diagnostic. A nonzero quadrature error is therefore distinct from an incorrect occupation count. The calculations do not renormalize the model density.

The Gaussian and plane-wave (GPW) and Gaussian and augmented-plane-wave (GAPW) methods provide the native density representations. Their exchange–correlation (XC) fields enter Skala. GAPW-XC denotes the XC-specific augmented density decomposition, and not merely the name of an input switch. All-electron (AE) GAPW is denoted GAPW-AE. Goedecker–Teter–Hutter (GTH) effective-core potentials (ECPs) define the pseudopotential Hamiltonian. GTH-direct and GTH-1C denote GAPW-GTH with direct valence and one-centre (1C) reconstructed fields, respectively. The mixed molecular and solid protocols combine AE and GTH treatments. They are not hybrid XC functionals. AE and GAPW-XC already contain their intrinsic one-centre terms. The independent with/without comparison applies to the two ordinary pseudopotential GAPW routes. Compact tables use GX for GAPW-XC/GTH, AE for GAPW-AE, and HD/HOC for the mixed AE/GTH protocols with direct/1C GTH fields, respectively.

Machine-readable reaction values, selected equation-of-state (EOS) curves, molecular-crystal total energies, and exact inputs are organized separately in the accompanying benchmark repository. The numerical tables below are generated from those datasets rather than transcribed from working logs.

S2 Reconstruction and Derivative Consistency

S2.1 Spin-gradient cross terms

Write the two reconstructed gradient fields as $\mathbf{g}_\alpha = \tilde{\mathbf{g}}_\alpha + \mathbf{g}_\alpha^h - \mathbf{g}_\alpha^s$ and analogously for β . Their three spin invariants are $\gamma_{\alpha\alpha} = \mathbf{g}_\alpha^2$, $\gamma_{\beta\beta} = \mathbf{g}_\beta^2$, and $\gamma_{\alpha\beta} = \mathbf{g}_\alpha \cdot \mathbf{g}_\beta$. For example,

$$\begin{aligned} \gamma_{\alpha\beta} = & \tilde{\mathbf{g}}_\alpha \cdot \tilde{\mathbf{g}}_\beta + \mathbf{g}_\alpha^h \cdot \mathbf{g}_\beta^h + \mathbf{g}_\alpha^s \cdot \mathbf{g}_\beta^s \\ & + \tilde{\mathbf{g}}_\alpha \cdot \mathbf{g}_\beta^h + \mathbf{g}_\alpha^h \cdot \tilde{\mathbf{g}}_\beta - \tilde{\mathbf{g}}_\alpha \cdot \mathbf{g}_\beta^s - \mathbf{g}_\alpha^s \cdot \tilde{\mathbf{g}}_\beta \\ & - \mathbf{g}_\alpha^h \cdot \mathbf{g}_\beta^s - \mathbf{g}_\alpha^s \cdot \mathbf{g}_\beta^h. \end{aligned} \quad (\text{S1})$$

The six component vectors admit $\binom{6}{2} = 15$ unordered off-diagonal dot products. Three are opposite-spin products within the same smooth, hard, or soft component, while twelve couple different components. Thus “15 cross terms” counts all off-diagonal pairs, not fifteen terms all absent from three independent spin-resolved evaluations. Separate smooth/hard/soft evaluations also combine diagonal terms with the wrong nonlinear structure. Reconstructing the fields first avoids this ambiguity and retains the nonlocal coupling of the model, which cannot be recovered by adding local algebraic corrections afterwards.

S2.2 The discrete adjoint and moving grids

For regular-grid values u_g and atom-grid interpolation coefficients I_{ig} ,

$$u_i = \sum_g I_{ig} u_g + u_i^h - u_i^s, \quad \frac{\partial E}{\partial u_g} = \sum_i I_{ig} \frac{\partial E}{\partial u_i}. \quad (\text{S2})$$

The reverse mapping uses the transpose of the forward interpolation, with the energy quadrature weights included consistently in the derivative. Hard and soft adjoints carry plus and minus signs. This applies separately to density, its Cartesian gradient components, and kinetic-energy density. At fixed orbital density matrix P , the explicit XC derivative for a nuclear coordinate or strain parameter λ includes both field and geometric terms,

$$\left. \frac{\partial E_{\text{xc}}}{\partial \lambda} \right|_P = \sum_i \left. \frac{\partial E_{\text{xc}}}{\partial u_i} \right|_P \frac{\partial u_i}{\partial \lambda} + \left. \frac{\partial E_{\text{xc}}}{\partial \lambda} \right|_u. \quad (\text{S3})$$

The last term encompasses explicit quadrature, partition, coordinate, and model-environment dependence. The total-energy force and stress additionally use electronic stationarity and the Gaussian-basis/overlap Pulay contributions, including cell and reciprocal-lattice dependence. The XC energy alone is not stationary in P . An adjoint test at fixed fields is necessary but does not replace a self-consistent total-energy finite-difference (FD) test.

S2.3 Finite-difference protocol

For a displaced coordinate $R_{A,x}$, the central FD estimate is

$$F_{A,x}^{\text{FD}}(h) = -\frac{E(R_{A,x} + h) - E(R_{A,x} - h)}{2h}. \quad (\text{S4})$$

For a homogeneous xx strain, both cell and Cartesian coordinates are transformed while fractional coordinates remain fixed. In the positive-pressure stress convention,

$$\sigma_{xx}^{\text{FD}}(\eta) = -\frac{E(+\eta) - E(-\eta)}{2\eta\Omega}. \quad (\text{S5})$$

At least two step sizes distinguish truncation error from SCF noise. An SCF threshold tightened for these checks does not change the production thresholds. A molecular coordinate-scaling virial is not a periodic stress-tensor validation, and energy agreement under symmetry reduction does not establish force equivalence after a symmetry-breaking displacement.

Self-consistent checks use one water molecule in a periodically repeated 8 Å cube at Gamma, without D3, to isolate the native electronic derivatives. The grid settings are 400/60 Ry, five multigrids, and 100 radial/434 angular points. The basis/core and reconstruction definitions match the corresponding production representations. The first hydrogen is displaced along x by $h = 10^{-3}$ and 5×10^{-4} bohr. Homogeneous xx strains are $\eta = 10^{-4}$ and 5×10^{-5} . Every perturbed calculation starts from the exact central wavefunction and converges independently to an OT residual below 10^{-9} . All 45 executions have normal termination and zero execution/time exits. The regular-grid dimensions remain unchanged under these perturbations.

Table S1: Signed FD minus analytic derivatives. Force errors are in $\mu E_h/\text{bohr}$ and stress errors in MPa. Columns give the two displacement/strain steps.

Representation	$\Delta F(10^{-3})$	$\Delta F(5 \times 10^{-4})$	$\Delta \sigma(10^{-4})$	$\Delta \sigma(5 \times 10^{-5})$
GPW/GTH	+0.228	+2.239	+0.038	-0.208
GAPW-XC/GTH	-0.377	+2.380	+0.012	-0.332
GAPW-AE	+2.855	-2.404	+0.007	-0.472
GAPW-GTH direct	+0.212	+1.414	+0.044	-0.185
GAPW-GTH one centre	-2.726	-1.977	-0.032	-0.017

These checks test the self-consistent total-energy derivative, including the reconstruction and moving-grid terms, rather than only model differentiation at fixed fields. The smaller steps do not uniformly reduce the error: subtraction of converged energies amplifies finite SCF and numerical noise. Consequently, no ideal quadratic step-size scaling is claimed. The test covers one Cartesian force component and one diagonal strain component in this periodic Gamma-point system. It is not a validation of every force component, shear strain, finite-k-point derivative, or symmetry-breaking displacement under point-group reduction. Exact inputs, outputs, timing records, and file/model/binary hashes accompany the machine-readable analysis.

S3 Molecular Numerical Settings and Convergence

S3.1 Basis, grid, and boundary conditions

The 70-reaction comparison uses the University of Zurich (UZH) 2026.2 AE TZVPP-MOLOPT-PBE-ae basis and corresponding explicit-electron Hamiltonian where available. The GTH part uses TZV2P-MOLOPT-PBE-GTH bases and GTH-PBE potentials.^{1,2} The all-GTH protocol uses GAPW-XC throughout. The mixed protocols share the light-element AE energies and use GTH for the retained heavy-element species, including Bi, Te, and I. This differs from the def2-based AE/ECP protocol in the molecular GauXC paper.

The molecule-centred cells add 25 Å to each molecular Cartesian extent. Cell and Poisson periodicity are disabled and the analytic isolated-system Poisson solver is used. The production atom-composite quadrature has 100 radial points and the 434-point Lebedev rule. Plane-wave/relative cutoffs are 400/60 Ry for GAPW-XC/GTH and 640/60 Ry for the shared AE route. The executed inputs document the corresponding GTH one-centre settings. OT and diagonalization thresholds are 5×10^{-7} and 5×10^{-6} , respectively. The threshold controls the solver residual, not an energy error in kcal mol^{-1} .

Basis/core changes must be kept distinct from quadrature refinement. The reported AE/GTH differences are not a basis-only convergence series, and no complete-basis limit is inferred from them. In the paired controls below, TZVPP and QZVPP denote the UZH triple- and quadruple-zeta valence plus double-polarization AE bases, respectively.

S3.2 Targeted paired all-electron basis controls

A targeted seven-reaction study was selected before calculation to sample covalent, hydrogen-bonded, dispersion-bound, and torsional energy differences. Five reaction pairs are complete in the present assessment. The W4-11/138 and WATER27/20 pairs remain incomplete. The controls retain the production geometries, charges, spin multiplicities, AE Hamiltonian, D3(BJ) convention, 640/60 Ry cutoffs, and 100/434 quadrature. Both basis levels use OT with threshold 5×10^{-7} . Thus the intended change is the AE orbital basis, not a change in the reference reaction or integration protocol. Mean absolute errors (MAEs) remain defined on the fixed production reaction set, not this targeted selection.

Table S2 reports every completed pair, including the unresolved QZVPP ethane result. The corresponding TZVPP reactions reproduce the archived production values to within 1.4×10^{-5} kcal mol⁻¹. Enlarging the basis reduces the CARBHB12/4 reference error from 5.61 to 0.03 kcal mol⁻¹ and the G2RC/15 error from 4.98 to 2.17 kcal mol⁻¹. The effects on ADIM6/2 and RG18/1 are much smaller and do not uniformly improve the reference agreement. The GauXC and PySCF columns use the same reaction-resolved source data as Sec. S4.1, retaining their own basis/core and dispersion protocols.

Table S2: Targeted paired molecular basis controls, in kcal mol⁻¹. Both bases use the same geometry, Hamiltonian, quadrature, and dispersion. The starred QZVPP ethane barrier is an unresolved outlier despite SCF convergence, not a recommended replacement. These selected reactions do not define a revised 70-reaction MAE.

Reaction	Reference	TZVPP	QZVPP	GauXC AE	PySCF
CARBHB12/4	9.967	15.577	9.939	10.388	10.369
ADIM6/2	1.990	2.088	2.321	1.852	1.860
BHROT27/1	2.730	2.836	28.009*	2.889	2.892
RG18/1	0.080	0.134	0.130	0.136	0.133
G2RC/15	-39.430	-44.408	-41.605	-36.070	-35.925

The QZVPP BHROT27/1 barrier of 28.009 kcal mol⁻¹ differs strongly from both the 2.730 reference and the 2.836 TZVPP result. Both conformers satisfy the SCF threshold, normal termination, and electron-count checks. Those checks alone do not establish that the desired electronic solution and numerical integration are adequate. The source energies are preserved, while the anomalous pair remains under electronic-state and quadrature investigation. It is not substituted into the main benchmark. No aggregate accuracy is inferred from this deliberately selected, incomplete basis study, and no favourable QZVPP replacements are mixed into the uniform 70-reaction dataset.

S3.3 Regular-grid cutoff

The paired conformational test uses the same two ACONF geometries and evaluates the relative energy within each representation. The 800 Ry result of that representation is the numerical reference. The small difference between the two 800 Ry relative energies is not set to zero: convergence within a representation and agreement between representations are separate questions.

Table S3: Conformational energy of the paired ACONF geometries in kcal mol⁻¹. Each error is relative to its own 800 Ry result. The relative cutoff is 60 Ry.

Cutoff/Ry	GPW	GAPW-XC	Δ GPW	Δ GAPW-XC
240	0.774857	0.726730	0.027401	-0.018666
280	0.704008	0.739410	-0.043448	-0.005985
320	0.747292	0.745360	-0.000165	-0.000035
360	0.746783	0.745610	-0.000673	0.000215
400	0.749608	0.745215	0.002152	-0.000181
480	0.750496	0.745307	0.003039	-0.000089
560	0.748038	0.745300	0.000581	-0.000095
640	0.747593	0.745297	0.000136	-0.000099
800	0.747457	0.745395	0.000000	0.000000

At 400 Ry, the deviations from the respective 800 Ry conformational energies are +0.002152 kcal mol⁻¹ for GPW and -0.000181 kcal mol⁻¹ for GAPW-XC. The faster GAPW-XC convergence supports separating hard local XC structure from the smooth auxiliary grid. This is an observable-specific cutoff test, distinct from comparison with an AE Hamiltonian. The complete inputs and outputs accompany the table.

Table S4: Self-consistent GPW water quadrature test at 800/60 Ry. Total-energy shifts are relative to 200/974 and are in microhartree. Electron-integral errors refer to eight explicit electrons.

Radial	Angular	$\Delta E/\mu E_h$	$\Delta N/e$
50	50	-369.201	+4.486e-04
50	100	+237.729	-4.318e-05
50	194	+263.762	-8.268e-06
50	302	+307.424	-1.074e-06
50	434	+245.349	-1.215e-07
100	50	-592.561	+4.491e-04
100	100	-10.739	-4.333e-05
100	302	+72.144	-8.919e-07
100	434	+8.379	-3.031e-07
100	590	+6.084	-2.206e-07
150	434	+12.627	-1.352e-07
150	590	+10.143	-2.079e-07
150	770	+14.433	-1.869e-07
200	590	+11.319	-2.275e-07
200	770	+15.822	-1.715e-07
200	974	+0.000	-1.347e-07

S3.4 Radial and angular quadrature

The water test refines radial and angular integration independently of the plane-wave representation. The electron integral is retained as a diagnostic, without rescaling the density. Neither the small water total-energy differences nor the conformational cutoff differences define a universal reaction-energy error bound. Frozen-density calculations that did not satisfy the stated SCF acceptance are not used as converged-energy evidence.

S3.5 Molecular cell padding

Selected matched isolated-system calculations tested added extents of 22, 24, and 25 Å. The maximum species-energy changes are 0.159, 0.189, and 0.348 μE_h for 22–24, 24–25, and 22–25 Å, respectively. The corresponding maximum reaction-energy changes are 0.000138, 0.000163, and 0.000298 kcal mol⁻¹. Maximum changes of the atom-grid electron integral are 7.94×10^{-8} , 1.43×10^{-7} , and 9.72×10^{-8} electrons. These comparisons retain the nonperiodic analytic electrostatic treatment. Old periodic or differently centred cells are not pooled with the final molecular data.

S4 Molecular Reaction Data and Cross-Code Comparisons

The common set contains 70 reactions from 36 parent subsets of dietGMTKN55.^{3,4} Each reaction has complete accepted species energies in every compared protocol. The remaining cases required longer SCF convergence. The final common set is fixed across methods. No additional reaction is removed on the basis of its reference error. The reaction index and source records in the repository define the selection explicitly.

The same D3 contribution is used in the native reaction comparison. Signed errors are calculation minus the official reference, and MAEs are unweighted averages over the fixed common set. They are not full-100 dietGMTKN55 statistics or type-2 weighted total mean absolute deviations (WTMAD-2). The ozone-containing W4-11/138 reaction remains included, despite its large GTH error. It contributes 58.8% of the all-GTH squared error. Removing it would lower the 69-reaction GAPW-XC MAE to 2.563 kcal mol⁻¹. This is an influence diagnostic, not the headline result. The direct/one-centre mixed comparison shares all light-element AE energies, and only one retained reaction changes by more than 10⁻⁹ kcal mol⁻¹.

Table S5: Complete common-set reaction energies in kcal mol⁻¹. Reference values are official benchmark energies. Protocol labels follow Sec. S1. No reference-error outlier is omitted.

Subset	Reaction	Ref.	GX	HD	HOC
ACONF	2	0.61400	0.30663	0.68210	0.68210
ACONF	5	0.59500	0.30757	0.58761	0.58761
ACONF	8	1.17800	0.77590	1.22152	1.22152
ADIM6	2	1.99000	2.72718	2.08787	2.08787
AHB21	1	-17.79000	-22.18751	-20.29619	-20.29619

Subset	Reaction	Ref.	GX	HD	HOC
AHB21	15	-8.62000	-11.59294	-10.86032	-10.86032
Amino20x4	23	4.04500	3.68879	4.42456	4.42456
Amino20x4	48	0.54000	0.27631	0.90052	0.90052
Amino20x4	56	1.88900	1.93478	1.58118	1.58118
BH76	6	17.80000	18.56504	17.86395	17.86395
BH76	18	2.50000	-8.63844	-1.27459	-1.27459
BH76RC	16	13.40000	10.89326	13.59788	13.59788
BHPERI	1	35.30000	33.09940	39.25014	39.25014
BHPERI	17	13.10000	10.81645	12.70129	12.70129
BHPERI	23	27.80000	28.29192	26.47568	26.47568
BHPERI	26	31.30000	31.40095	30.85649	30.85649
BHROT27	1	2.73000	2.69043	2.83649	2.83649
BHROT27	10	8.03000	8.43958	8.00707	8.00707
BSR36	4	8.85000	8.44034	8.81562	8.81562
BSR36	21	9.78000	8.34085	8.82335	8.82335
BUT14DIOL	28	2.83000	2.69982	4.33282	4.33282
BUT14DIOL	39	3.10000	3.11306	4.93629	4.93629
BUT14DIOL	63	4.31000	4.09728	5.97195	5.97195
CARBHB12	3	2.42100	2.69672	3.68798	3.68798
CARBHB12	4	9.96700	9.97430	15.57723	15.57723
CARBHB12	6	3.02100	2.95159	5.16892	5.16892
CARBHB12	11	3.24100	3.92467	3.06786	3.06786
CDIE20	2	4.40000	4.48598	4.99901	4.99901
CDIE20	14	4.40000	4.56044	4.54726	4.54726
CHB6	1	-34.43000	-33.49431	-34.18723	-34.18723
DC13	6	-25.70000	-32.03193	-31.81763	-31.81763
DC13	12	-58.70000	-60.73559	-67.15589	-67.15589
FH51	25	-20.27000	-21.62745	-19.42107	-19.42107
G21EA	5	17.30000	-1.42565	10.29928	10.29928
G21EA	25	54.70000	50.95114	51.72694	51.72694
G21IP	28	243.70900	239.85434	243.88740	243.88740
G2RC	15	-39.43000	-40.17337	-44.40802	-44.40802
HAL59	21	10.54000	12.10722	9.89950	9.89950
HEAVY28	5	0.98000	2.32113	1.35292	1.35483
ICONF	16	3.66000	4.07657	3.99415	3.99415
IL16	3	-116.91000	-116.64846	-119.94685	-119.94685
INV24	12	10.30000	11.08966	10.79061	10.79061
INV24	19	6.20000	5.46217	6.10591	6.10591
INV24	22	27.20000	27.35357	27.74964	27.74964
ISO34	12	45.65000	48.13511	45.85998	45.85998
ISOL24	13	33.02000	34.47575	34.65021	34.65021
MB16-43	4	356.84810	320.43581	358.96453	358.96453
MCONF	7	2.92000	3.11166	2.61484	2.61484
MCONF	22	5.31000	5.83140	4.73929	4.73929
MCONF	30	4.86000	5.03309	5.16786	5.16786
MCONF	42	7.32000	8.16255	7.33699	7.33699
PArel	3	2.94000	2.90267	3.05754	3.05754
PArel	5	7.19000	8.68203	7.75014	7.75014
PArel	10	2.98000	2.64094	2.92259	2.92259
PArel	20	2.15000	1.17708	1.12222	1.12222
PNICO23	13	3.98000	4.16981	4.46224	4.46224
PNICO23	17	7.10000	7.39778	7.31551	7.31551
RG18	1	0.08000	0.17488	0.13391	0.13391
RG18	15	0.24000	0.48594	0.28332	0.28332
S66	11	5.42000	5.85282	5.96275	5.96275
S66	17	17.18000	17.67587	19.17660	19.17660
S66	24	2.82000	3.50720	3.46501	3.46501
S66	45	1.75000	2.59070	2.03582	2.03582
SCONF	10	5.65000	6.09695	8.67396	8.67396
SIE4x4	7	31.30000	45.77575	51.75030	51.75030
W4-11	136	126.38500	92.90307	129.75705	129.75705
W4-11	138	147.42800	78.94035	156.69640	156.69640
WATER27	17	76.50000	76.32827	82.77137	82.77137

Subset	Reaction	Ref.	GX	HD	HOC
WATER27	20	26.60000	29.31104	33.76403	33.76403
YBDE18	1	57.17000	55.99377	60.22366	60.22366

S4.1 Direct comparison with the previous molecular implementation

All PySCF/Skala reference values use Becke partitioning without an atomic-radius adjustment.

The source is the reaction-resolved collaborator dataset archived with the earlier molecular GauXC study, at repository revision 25e80ba.^{5,6} Each native reaction is matched by its species geometries after translation, charge, multiplicity, and signed stoichiometric coefficients. Official references and weights are checked independently. All 70 retained reactions have complete GauXC AE/ECP and PySCF energies. Converged GauXC GPW-GTH energies cover 65 of them. Source reaction numbers differ from the native benchmark identifiers, and the deposited mapping resolves those aliases explicitly.

The comparisons are separated by core treatment. Native GAPW-XC/GTH is compared with GauXC GPW-GTH on the 65-reaction intersection. The mixed native AE/GTH protocols are compared with GauXC AE/def2-ECP and PySCF on the full 70-reaction common set. The earlier AE/ECP calculations use def2-TZVP/ma-def2-TZVP bases, with def2 ECPs for the heavier elements, rather than the native UZH basis/core protocol specified in Sec. S3.1. This is the principal protocol distinction: the finite Gaussian space changes and, where ECPs differ, so does the core Hamiltonian. The remaining reaction-energy differences are consistent with these basis/core and numerical integration choices. The earlier GPW-GTH calculations instead use a SCAN-optimized basis/pseudopotential combination, whereas the native GTH protocol uses its PBE-optimized counterpart. The comparison complements the numerical consistency tests in Secs. S2 and S3, which validate the native implementation.

The following tables abbreviate GauXC AE/def2-ECP as GauXC AE and GauXC GPW-GTH as GauXC GPW where space is limited. Mean absolute differences (MADs) refer to direct differences between calculated reaction energies, not differences of reference MAEs. Root-mean-square (RMS) errors and differences are reported separately.

Table S6: Reference errors on reaction-matched populations, in kcal mol⁻¹. The first five rows use the fixed native 70-reaction set. The last two use its 65-reaction intersection with converged GauXC GPW-GTH data. Each source retains its recorded dispersion contribution.

Method	<i>N</i>	MAE	Median absolute	RMS	Maximum absolute
Native GX	70	3.5051	0.6025	10.6749	68.4877
Native HD	70	1.8905	0.5549	3.6502	20.4503
Native HOC	70	1.8905	0.5549	3.6502	20.4503
GauXC AE/ECP	70	1.2378	0.2777	2.7842	14.9074
PySCF	70	1.2314	0.3048	2.7662	14.9606
Native GX	65	2.2193	0.4919	8.8172	68.4877
GauXC GPW-GTH	65	2.1811	0.5820	8.6673	67.4910

On the 65-reaction pseudopotential intersection, native GAPW-XC/GTH and GauXC GPW-GTH have reference MAEs of 2.2193 and 2.1811 kcal mol⁻¹. Their direct MAD is 0.4631 kcal mol⁻¹, with signed mean -0.2250, RMS difference 0.7799, and maximum absolute difference 3.9442 for DC13/6. By contrast, on all 70 reactions the mixed native protocols have reference MAEs of 1.8905, compared with 1.2378 for GauXC AE/ECP and 1.2314 for PySCF. The Mixed D-GauXC AE/ECP MAD is 1.8889 kcal mol⁻¹, with signed mean +0.6494 and median absolute difference 0.7151. The largest difference, +18.6055 kcal mol⁻¹ for W4-11/138, remains included. No outlier is removed from either intersection.

Table S7: Direct reaction-energy differences, in kcal mol⁻¹. The first three difference statistics use the recorded total energies. The last column removes each source's own D3 contribution to isolate its influence on the comparison.

Method pair	<i>N</i>	MAD	RMS	Maximum	Electronic MAD
Native GX / GauXC GPW-GTH	65	0.4631	0.7799	3.9442	0.4624
Native HD / GauXC AE/ECP	70	1.8889	3.5718	18.6055	1.8880
Native HOC / GauXC AE/ECP	70	1.8889	3.5718	18.6055	1.8880
Native HD / PySCF	70	1.8925	3.5585	18.3927	1.8857
GauXC AE/ECP / PySCF	70	0.0456	0.0787	0.4059	0.0400

All headline comparisons retain the total energies as recorded, including each source's separately evaluated D3(BJ) correction. The source calculations use the B3LYP5 dispersion convention, but the recorded corrections are not numerically identical across every protocol. For native Mixed D versus GauXC AE/ECP, the D3 reaction-contribution

differences have a MAD of only 0.001396 kcal mol⁻¹ and a maximum of 0.014549. Removing each source's own correction changes the direct MAD from 1.8889 to 1.8880 kcal mol⁻¹. Dispersion bookkeeping therefore does not explain the larger native-GauXC differences. For native GAPW-XC/GTH versus GauXC GPW-GTH, the corresponding MAD changes from 0.4631 to 0.4624. The latter reproduces the earlier comparison in which the same native D3 term was added to both electronic reaction energies.

There is one non-negligible recorded GauXC AE/ECP-PySCF dispersion mismatch within the retained set: HEAVY28/5 differs by +0.419041 kcal mol⁻¹ in its D3 reaction contribution. Its origin is not established, and neither source value is overwritten. Across the 70 reactions, the GauXC-PySCF direct MAD is 0.04560 for the recorded totals and 0.03999 for the electronic contributions alone. Their total-energy differences should therefore not all be attributed to quadrature. The comparison files preserve electronic, dispersion, and total energies separately, together with input/output hashes and the pinned source revision.

Table S8: Reaction-resolved collaborator energies on the native common set, in kcal mol⁻¹. Reaction identifiers and official references match Table S5. A dash denotes an unavailable converged GPW-GTH value, not a zero. Recorded dispersion contributions are included.

Subset	Reaction	Ref.	GauXC GPW	GauXC AE	PySCF
ACONF	2	0.61400	0.36237	0.64648	0.68837
ACONF	5	0.59500	0.31848	0.55307	0.52619
ACONF	8	1.17800	0.77859	1.18813	1.15144
ADIM6	2	1.99000	2.30458	1.85184	1.85984
AHB21	1	-17.79000	-20.45402	-17.39888	-17.40971
AHB21	15	-8.62000	-11.71918	-8.79444	-8.79323
Amino20x4	23	4.04500	3.85348	3.71127	3.63208
Amino20x4	48	0.54000	0.01575	0.60110	0.61964
Amino20x4	56	1.88900	2.20238	2.03964	2.03641
BH76	6	17.80000	19.11753	17.03316	17.15220
BH76	18	2.50000	—	-0.21392	-0.22005
BH76RC	16	13.40000	10.99734	14.26280	14.25226
BHPERI	1	35.30000	34.24032	36.03270	36.07884
BHPERI	17	13.10000	11.23120	12.99901	13.04625
BHPERI	23	27.80000	28.41555	28.73250	28.79864
BHPERI	26	31.30000	31.41251	31.55939	31.44178
BHROT27	1	2.73000	2.83023	2.88882	2.89208
BHROT27	10	8.03000	8.43419	8.32717	8.32723
BSR36	4	8.85000	8.73777	8.15235	8.24610
BSR36	21	9.78000	9.14667	7.99739	8.10612
BUT14DIOL	28	2.83000	2.76824	2.95418	2.96389
BUT14DIOL	39	3.10000	3.33710	3.62532	3.65624
BUT14DIOL	63	4.31000	4.08891	4.48423	4.52609
CARBHB12	3	2.42100	2.64009	2.55364	2.56030
CARBHB12	4	9.96700	9.78532	10.38773	10.36877
CARBHB12	6	3.02100	2.80568	2.36925	2.38662
CARBHB12	11	3.24100	3.96770	3.39157	3.39480
CDIE20	2	4.40000	4.52498	4.36188	4.36983
CDIE20	14	4.40000	4.80957	4.23473	4.22647
CHB6	1	-34.43000	-32.06178	-35.71422	-35.72727
DC13	6	-25.70000	-28.08768	-32.02494	-32.00898
DC13	12	-58.70000	-62.03843	-63.80513	-63.59949
FH51	25	-20.27000	-22.33482	-21.14074	-21.08624
G21EA	5	17.30000	—	17.43069	17.44739
G21EA	25	54.70000	52.40220	56.66247	56.67548
G21IP	28	243.70900	240.91373	246.10544	246.06756
G2RC	15	-39.43000	-40.58825	-36.07029	-35.92457
HAL59	21	10.54000	13.16196	10.18052	10.22753
HEAVY28	5	0.98000	—	1.01418	0.60831
ICONF	16	3.66000	4.21525	3.91693	3.89953
IL16	3	-116.91000	-116.14007	-117.17430	-117.26645
INV24	12	10.30000	11.29756	10.42813	10.40776
INV24	19	6.20000	6.15822	5.90506	5.91207
INV24	22	27.20000	28.71502	27.03272	27.02526
ISO34	12	45.65000	48.12960	44.81214	44.83039

Subset	Reaction	Ref.	GauXC GPW	GauXC AE	PySCF
ISOL24	13	33.02000	34.15054	32.82246	32.75595
MB16-43	4	356.84810	–	349.30785	349.27767
MCONF	7	2.92000	3.40584	2.76081	2.75618
MCONF	22	5.31000	5.98878	5.70352	5.65919
MCONF	30	4.86000	5.02760	5.10589	5.07451
MCONF	42	7.32000	8.08468	7.86694	7.84842
PArel	3	2.94000	3.04128	2.97323	2.95625
PArel	5	7.19000	9.07152	7.36215	7.35467
PArel	10	2.98000	2.86582	3.35368	3.35918
PArel	20	2.15000	1.55954	1.95014	2.02341
PNICO23	13	3.98000	4.24282	4.57238	4.57621
PNICO23	17	7.10000	7.50137	7.35457	7.39182
RG18	1	0.08000	0.12640	0.13584	0.13272
RG18	15	0.24000	0.34321	0.48691	0.49133
S66	11	5.42000	5.52785	5.65511	5.66394
S66	17	17.18000	16.58982	17.11875	17.10879
S66	24	2.82000	2.91409	3.02703	3.01553
S66	45	1.75000	2.48973	1.89424	1.90961
SCONF	10	5.65000	6.23204	6.22279	6.19781
SIE4x4	7	31.30000	45.74439	46.20737	46.26060
W4-11	136	126.38500	–	119.95421	120.14384
W4-11	138	147.42800	79.93703	138.09093	138.30367
WATER27	17	76.50000	76.34800	81.45738	81.39384
WATER27	20	26.60000	30.45634	26.30882	26.24992
YBDE18	1	57.17000	54.63500	58.85771	58.76963

For context, the prior GauXC manuscript’s updated primary AE/ECP–PySCF comparison comprises all 100 reactions, not these 65 or 70.⁵ The values in Table S9 are from that updated analysis and are consistent with the reaction-resolved source supplied on 11 September 2026. They supersede earlier draft aggregates and differ from the first arXiv version. They must not be compared with the native aggregates as though they used the same reaction population.

Table S9: Context from the updated 100-reaction molecular GauXC manuscript. These aggregates are quoted for their original set, not rescored or compared as if they belonged to the native 70-reaction set. Energies are in kcal mol^{−1}.

Metric	CP2K/GauXC AE/ECP	Microsoft PySCF
MAE against reference	1.255	1.243
WTMAD-2	3.486	3.459

Direct cross-code MAD: 0.058; signed-error R^2 : 0.998; all absolute cross-code residuals below 0.5.

The difference of the 100-reaction aggregate MAEs, 0.012 kcal mol^{−1}, is not the direct cross-code MAD of 0.058 kcal mol^{−1}. Nor can either quantify native/GauXC equivalence under a changed basis and core treatment. Tables S6–S8 instead provide the explicit common-set comparison directly from the deposited species energies.

S5 K-Point and Symmetry Checks

Symmetry reduction is a change in the quadrature representation of a fixed Brillouin-zone integral. Correct weights, rotations of orbital components, and Bloch-phase conventions are required. For FD forces, the perturbed geometry may have lower symmetry than the reference geometry. Consistent unreduced sampling or correctly regenerated symmetry is needed. Inversion/time-reversal reduction and full point-group reduction are not interchangeable validation labels.

Seven archived controls compare complete and reduced 4³ meshes for C and MgO at unchanged geometry and scientific input. The full point-group reduction uses 10 irreducible points from 64, at symmetry tolerance 10^{−8}. Table S10 lists the absolute energy difference per reported cell. These controls establish agreement for their specific geometries, meshes, and executables. They are not a proof for all future restarts or all symmetry operations.

The C inputs of both mixed protocols use an AE carbon basis and potential, so those rows do not constitute additional GTH validations. Mixed MgO uses GTH Mg and AE O. The molecular-crystal production uses full point-group reduction of 3³ meshes. The independently reported CO₂ 5³ and NH₃ tolerance checks below constrain sampling and approximate-symmetry sensitivity for those cases. They should not be relabelled as full-versus-reduced tests.

Table S10: Selected full/reduced-grid energy checks. Residuals are the final printed SCF values. $|\Delta E|$ is in kJ mol⁻¹ of simulation cells.

System	Representation	Full SCF residual	Reduced SCF residual	$ \Delta E $
C	GAPW-XC/GTH	3.0×10^{-7}	2.1×10^{-7}	0.0200
C	GAPW-AE	8.0×10^{-8}	9.0×10^{-8}	0.0229
C	Mixed direct (AE here)	8.0×10^{-8}	9.0×10^{-8}	0.0229
C	Mixed one centre (AE here)	8.0×10^{-8}	9.0×10^{-8}	0.0229
MgO	GAPW-XC/GTH	4.7×10^{-7}	3.6×10^{-7}	0.0291
MgO	Mixed direct	2.7×10^{-7}	4.9×10^{-7}	0.0493
MgO	Mixed one centre	1.4×10^{-7}	9.0×10^{-8}	0.0378

S6 Molecular-Crystal Energies and Numerical Controls

S6.1 Geometries and electronic protocol

The crystal and gas-phase geometries are taken from Sec. 5.6 of the versioned supporting dataset arXiv:2402.13059v1 of Della Pia *et al.*⁷ Crystal coordinates are wrapped into the supplied cells without relaxation or symmetrization. CO₂ and NH₃ have cubic cells with $a = 5.624$ and 5.1305 Å and four molecules each. Urea has a tetragonal cell with $a = b = 5.565$, $c = 4.684$ Å, and two molecules. The separately supplied gas molecules are rigidly translated into centred cells whose dimensions are the molecular extents plus 25 Å. They use nonperiodic cell/Poisson settings and the analytic isolated-system solver.

All four representations use Skala-1.1 Rev1 on a smoothly partitioned atom-composite quadrature. The baseline quadrature has 150 radial points and the 770-point Lebedev rule. Five multigrids use cutoffs of 800/60 Ry in both phases. The baseline AE calculations use the UZH 2026.2 TZVPP-MOLOPT-PBE-ae basis and explicit nuclei. All three reported AE pairs instead use QZVPP-MOLOPT-PBE-ae with 200 radial points and the 974-point Lebedev rule, following the matched tests in Sec. S6.3. GTH calculations retain TZV2P-MOLOPT-PBE-GTH bases, GTH-PBE potentials, and the baseline quadrature. One-centre calculations employ the small extended expansion and accurate augmented-XC integration. The AE hard-minus-soft fields are reconstructed before the joint model evaluation. Direct and reconstructed GTH calculations share their Hamiltonian and basis.

D3(BJ) is included once, with $s_6 = 1$, $s_8 = 1.9889$, $a_1 = 0.3981$, $a_2 = 4.4211$, and interaction cutoff 20 in the executed input convention.^{8,9} The B3LYP parameter alias selects these damping constants, not the electronic functional. All systems are neutral restricted singlets. OT convergence is 5×10^{-7} . Complex k-point orbitals use iterative refinement of the approximative congruence transformation with the full AE preconditioner. No density renormalization, energy alignment, or fitting to the reference lattice energies is applied.

Every crystal uses a Gamma-centred 3^3 mesh with full point-group reduction. With symmetry tolerance 10^{-6} , baseline AE and GAPW-XC give 4, 10, and 10 irreducible points for CO₂, NH₃, and urea. The matched fine-grid CO₂ and NH₃ AE basis controls use 10^{-4} and four irreducible points in both bases, while urea retains ten. The GTH-direct and GTH-1C calculations use 10^{-4} and give 4, 4, and 6. These finite tolerances recognize near-symmetries in the decimal coordinates without moving the atoms. Their numerical sensitivity is distinct from a full-versus-reduced-grid comparison.

S6.2 Paired energies and controls

Table S11 gives all paired total energies. We use $E_{\text{latt}} = E_{\text{solid}}/Z - E_{\text{molecule}}$ and $1 E_h = 2625.4996394799$ kJ mol⁻¹. The electronic diffusion Monte Carlo (DMC) references have statistical uncertainties of 0.2, 0.1, and 0.3 kJ mol⁻¹ for CO₂, NH₃, and urea. Their estimated overall uncertainty is about 2 kJ mol⁻¹, rather than the smaller statistical bars alone.⁷ Thermal and zero-point corrections to experimental sublimation enthalpies are not added to the DMC energies.

Device comparisons distinguish central processing unit (CPU) inference from accelerator inference through CUDA. The CPU/CUDA labels refer only to this device choice.

For NH₃, CO₂, and urea, respectively, the reported absolute AE errors are 0.015, 2.667, and 1.927 kJ mol⁻¹, or 0.04%, 9.1%, and 1.8% of the DMC binding magnitudes. GAPW-XC/GTH gives 1.874, 8.008, and 5.225 kJ mol⁻¹, or 4.9%, 27.2%, and 4.8%. Thus NH₃ has the smallest absolute error and CO₂ the largest absolute and fractional error in each reported representation. These errors describe the combined Skala-D3(BJ) protocol at the stated basis and geometry, not separate XC, dispersion, or deformation-energy components.

Refining the paired CO₂ quadrature from 150/770 to 200/974 changes E_{solid}/Z by -0.110687 and E_{molecule} by -0.155241 kJ mol⁻¹. Their difference, $+0.044554$ kJ mol⁻¹, measures the lattice-energy sensitivity. The 5^3 crystal check uses 11 of 125 irreducible points, compared with four of 27 at 3^3 . For k-point, symmetry-tolerance, and device checks the molecule is unchanged, so the crystal shift per molecule is also the lattice-energy shift.

Table S11: Accepted total energies including D3(BJ), in hartree, and paired lattice energies in kJ mol^{-1} . Crystal energies refer to the complete cell. Extra digits support reproducibility and are not an accuracy claim.

System	Representation	Z	E_{solid}	E_{molecule}	E_{latt}
CO ₂	GAPW-XC-GTH	4	-151.045238668728	-37.747061778360	-37.407827
CO ₂	GAPW-AE	4	-754.437615342695	-188.597190183387	-32.066940
CO ₂	GAPW-GTH, direct	4	-151.045104012149	-37.747040234695	-37.376005
CO ₂	GAPW-GTH, one center	4	-151.045343896582	-37.747094149196	-37.391906
NH ₃	GAPW-XC-GTH	4	-47.028180200438	-11.741781495152	-40.074458
NH ₃	GAPW-AE	4	-226.314440915110	-56.564066501520	-38.184551
NH ₃	GAPW-GTH, direct	4	-47.028284463885	-11.741799977614	-40.094368
NH ₃	GAPW-GTH, one center	4	-47.028166197256	-11.741777335794	-40.076187
Urea	GAPW-XC-GTH	2	-88.087642387844	-44.000505611404	-113.725046
Urea	GAPW-AE	2	-450.662994683326	-225.289437740814	-110.427467
Urea	GAPW-GTH, direct	2	-88.087842193572	-44.000575209028	-113.804613
Urea	GAPW-GTH, one center	2	-88.087646982008	-44.000510082308	-113.719339

Table S12: Bounded numerical sensitivities in kJ mol^{-1} per molecule. All unlisted scientific settings are held fixed. Seven accepted control executions produce six comparisons because the atom-grid refinement requires both phases.

System and representation	Change	Observable	Shift
CO ₂ , GAPW-XC-GTH	150/770 to 200/974	ΔE_{latt} , both phases	+0.044554
CO ₂ , GAPW-XC-GTH	800 to 1200 Ry	$\Delta(E_{\text{solid}}/Z)$ only	-0.015499
CO ₂ , GAPW-XC-GTH	3 ³ to 5 ³	ΔE_{latt}	+0.000391
NH ₃ , GAPW-XC-GTH	Symmetry tolerance 10 ⁻⁶ to 10 ⁻⁴	ΔE_{latt}	-0.000429
CO ₂ , GAPW-XC-GTH	CPU minus CUDA	ΔE_{latt}	-0.000127
Urea, GAPW-AE	CPU minus CUDA	ΔE_{latt}	-0.000379

The accepted 1200 Ry cutoff result concerns only the CO₂ crystal and is not a paired binding-energy convergence test. The separate gas-phase calculation lacks a complete runtime-provenance record and is not used. CO₂ CPU/CUDA and urea-AE CPU/CUDA controls retain the respective executable and scientific input apart from model device selection. Their sub-0.001 kJ mol^{-1} changes constrain device effects for these cases. No complete-basis limit, basis-set superposition error (BSSE) correction, equilibrium volume, elastic response, or vibrational thermodynamics is inferred from the fixed-geometry energies.

S6.3 Paired all-electron basis sensitivity

The three basis tests use a matched crystal and gas molecule at each level. Within each pair of basis levels, the CPU model, geometry, AE Hamiltonian, one-centre reconstruction, dispersion parameters, 800/60 Ry cutoffs, 200/974 quadrature, k-point reduction, and SCF threshold are unchanged. The Urea series additionally separates the earlier quadrature refinement from basis enlargement. No geometry relaxation or reference-energy adjustment is involved.

Table S13: Paired AE basis and quadrature checks. Total energies are in hartree and lattice energies in kJ mol^{-1} . Both phases use the same basis and grid. All twelve fine-grid executions satisfy the OT threshold 5×10^{-7} and end normally. The QZVPP pairs supply all three AE entries in Table S11.

System	Basis	Radial/Lebedev	E_{solid}	E_{molecule}	E_{latt}
CO ₂	TZVPP	200/974	-754.203788644477	-188.537092135137	-36.376366
CO ₂	QZVPP	200/974	-754.437615342695	-188.597190183387	-32.066940
NH ₃	TZVPP	200/974	-226.289321267777	-56.557573718526	-38.743444
NH ₃	QZVPP	200/974	-226.314440915110	-56.564066501520	-38.184551
Urea	TZVPP	150/770	-450.589911695840	-225.247483909549	-124.637557
Urea	TZVPP	200/974	-450.590056186809	-225.247477953622	-124.842875
Urea	QZVPP	200/974	-450.662994683326	-225.289437740814	-110.427467

At the triple-zeta basis, changing the paired quadrature from 150/770 to 200/974 shifts the lattice energy from -124.637557 to -124.842875 kJ mol^{-1} , a change of -0.205318 kJ mol^{-1} . Enlarging both phase bases to quadruple-zeta at 200/974 instead gives -110.427467 kJ mol^{-1} , a change of +14.415408 kJ mol^{-1} . The initial large AE binding energy

is therefore substantially basis sensitive rather than a consequence of incomplete SCF convergence. The quadruple-zeta molecule and crystal converge in 114 and 29 steps, respectively, with final residuals 4.6×10^{-7} and 3.4×10^{-7} . The reported larger-basis value differs from DMC by $-1.927467 \text{ kJ mol}^{-1}$ and from the three GTH results by approximately $+3.3 \text{ kJ mol}^{-1}$.

The same basis enlargement changes CO_2 from -36.376366 to $-32.066940 \text{ kJ mol}^{-1}$ and NH_3 from -38.743444 to $-38.184551 \text{ kJ mol}^{-1}$. The corresponding shifts are $+4.309426$ and $+0.558893 \text{ kJ mol}^{-1}$. CO_2 has a substantial basis contribution, while ammonia changes less and approaches the DMC value within its statistical uncertainty. The QZVPP crystal/molecule executions converge in 47/88 steps for CO_2 and 60/35 for NH_3 . All final residuals are at or below 5×10^{-7} . The eight additional executions complete the same basis comparison for all three systems, rather than selecting a larger basis only where it improves a reference error.

An AE basis must resolve tightly localized core orbitals as well as the valence region, whereas the GTH Hamiltonian replaces the core electrons. Equal nominal zeta labels consequently do not establish equal completeness across the two core treatments. For a lattice energy, however, the decisive quantity is the differential basis error between crystal and molecule. The present basis enlargement changes that balance and can also alter BSSE. It does not isolate a core-only contribution or establish a complete-basis or counterpoise-corrected limit. Basis selection follows the paired numerical sensitivity, not removal of a system or adjustment of the functional to the DMC value.

S6.4 Other density-functional approximations

Table S14 compares the native results with the DMC references and published electronic lattice energies collected by Della Pia *et al.*^{7,10} Their plane-wave calculations use hard projector augmented-wave (PAW) datasets, a 1000 eV cutoff, and dense system-specific k-point meshes. The PBE0+MBD values use FHI-aims. These are literature protocols, not same-basis native-code controls or Skala reruns. The MAE is the unweighted mean of $|E_{\text{latt}} - E_{\text{latt}}^{\text{DMC}}|$ over CO_2 , NH_3 , and urea, using unrounded native energies and the reported literature values.

Table S14: Electronic lattice energies and three-crystal MAEs, all in kJ mol^{-1} . DMC references are from Ref. 7, with statistical uncertainties in parentheses. Native Skala-D3(BJ) values are from Table S11. Other density-functional values are transcribed from Tables S5, S8, and S22 of Ref. 10. Each dispersion model is part of the stated method. MAEs summarize only these three crystals and are not full-X23 statistics. Different geometrical and numerical protocols preclude interpreting the differences as implementation errors or a definitive method ranking.

Method	CO_2	NH_3	Urea	MAE
DMC reference	-29.4(2)	-38.2(1)	-108.5(3)	–
Skala-D3(BJ), GAPW-XC/GTH	-37.41	-40.07	-113.73	5.04
Skala-D3(BJ), GAPW-AE	-32.07	-38.18	-110.43	1.54
Skala-D3(BJ), GAPW-GTH direct	-37.38	-40.09	-113.80	5.06
Skala-D3(BJ), GAPW-GTH one-centre	-37.39	-40.08	-113.72	5.03
PBE	-6.51	-28.24	-73.24	22.70
PBE+D3	-25.07	-43.76	-109.90	3.76
PBE+MBD	-22.74	-42.10	-109.11	3.72
revPBE+D3	-22.60	-38.32	-99.88	5.18
SCAN+rVV10	-31.47	-41.69	-113.55	3.54
vdW-DF2	-32.75	-39.70	-103.85	3.17
PBE0+MBD (FHI-aims)	-23.74	-38.99	-109.40	2.45

The native pseudopotential MAEs of 5.03 – 5.06 kJ mol^{-1} are close to revPBE+D3 (5.18), although several other dispersion-inclusive approximations have smaller three-crystal MAEs. The AE MAE is 1.54 kJ mol^{-1} with all three QZVPP pairs. NH_3 is accurate in both native core treatments, the QZVPP AE urea residual is 1.93 kJ mol^{-1} , and pseudopotential urea errors are close to SCAN+rVV10 and vdW-DF2. CO_2 retains the largest AE residual, reduced to 2.67 kJ mol^{-1} by basis enlargement. The approximately 2 kJ mol^{-1} overall DMC uncertainty and the differing computational protocols caution against ranking methods by small residual differences.⁷ The results support promising, material-dependent condensed-phase transfer and substantial basis sensitivity, rather than universal strong overbinding.

The accompanying benchmark directory contains six source geometries, the 24 original base inputs, twelve additional paired AE basis/grid-control inputs, accepted executed inputs, and compact energy and provenance records. Original triple-zeta results remain archived separately from the selected QZVPP pairs. Calculations use CP2K revision 37aedacbb3, with full source and host-specific executable hashes in the archive. The crystal energies and numerical-control results are retained separately.

S7 Preliminary Ice Basis Comparison

Six phases from DMC-ICE13 currently have completed AE calculations with both orbital bases: VI, VII, VIII, IX, XV, and XVII. Their electronic lattice energies provide a second test of the basis sensitivity observed for the molecular crystals. The DMC references are from Della Pia *et al.*¹¹ The present table is a fixed six-phase assessment, not the completed 13-phase QZVPP benchmark or a phase-stability ranking relative to ice Ih.

Both basis levels use the same fixed crystal geometries, Skala–D3(BJ), explicit-electron Hamiltonian, joint one-centre reconstruction, 800/60 Ry cutoffs, and 150/770 atom quadrature. The SCF threshold is 5×10^{-7} with OT. The QZVPP crystals use Gamma-centred 3^3 meshes with full point-group reduction through spglib, yielding 14, 8, 6, 6, 14, and 14 irreducible points for VI, VII, VIII, IX, XV, and XVII, respectively. All executions use one process and 32 CPU threads on ROSI. Exact executable, input, model, timing, and output records accompany the energies.

The earlier TZVPP calculations use CP2K’s K290 symmetry backend, including its inversion fallback for ice IX. Separate same-basis spglib checks for IX and VII change the energy by -0.000035 and $+0.000200$ kJ mol⁻¹ per molecule, respectively. These controls distinguish the symmetry-backend change from the much larger basis shifts. No inversion fallback occurs in the six reported QZVPP runs.

The gas-phase reference is a centred isolated water molecule at the reconstructed Partridge–Schwenke equilibrium geometry, with OH distance 0.95865 Å and HOH angle 104.348°.¹² Its total energies are -76.412709030115 and -76.436220026726 hartree at TZVPP and QZVPP, respectively. This geometrical convention follows the equilibrium molecular reference, but the original DMC monomer coordinate file was not available for a direct file-level comparison. Any molecular-reference mismatch would shift all absolute ice lattice energies together. Relative phase energies cancel that molecular reference, but are not reported here while the corresponding QZVPP Ih reference remains incomplete.

Table S15: Preliminary six-phase ice basis comparison. Electronic lattice energies and signed QZVPP–DMC residuals are in kJ mol⁻¹ per water molecule. Parentheses give DMC statistical uncertainties. The averages concern these same six phases only, not the complete DMC-ICE13 benchmark.

Phase	TZVPP	QZVPP	DMC	QZVPP–DMC
VI	-80.663	-56.275	-57.67(7)	+1.395
VII	-80.495	-55.128	-54.46(7)	-0.668
VIII	-79.642	-54.803	-55.22(8)	+0.417
IX	-78.841	-57.209	-58.85(7)	+1.641
XV	-80.474	-56.286	-57.71(7)	+1.424
XVII	-75.684	-56.674	-57.70(8)	+1.026
MAE	22.365	1.095	–	–

On these same six phases, the MAE decreases from 22.365 to 1.095 kJ mol⁻¹. Five QZVPP energies are slightly less binding than DMC and one is more binding. The large TZVPP overbinding is therefore not retained with the enlarged AE basis and fully reduced k-point protocol. This preliminary result identifies numerical basis sensitivity as essential to the interpretation, rather than establishing a complete-basis limit or a universal accuracy claim for all ice phases. No empirical energy shift is applied. The full crystal total energies, molecular normalization, and two symmetry controls can be reproduced from the accompanying basis-assessment script.

S8 Ten-Solid Equations of State

S8.1 Structures and common fit windows

The selected solids are AlN, AlP, BN, BP, C, LiCl, MgO, MgS, Si, and SiC from the Goldzak set.¹³ All use conventional cubic eight-atom cells. Ten initial volumes are evenly spaced from 0.90 to 1.10 times the static-lattice experimental volume. The common fit window contains the first nine volumes for AlN and AlP, ending at 1.0777778, and all ten for the other solids. The 40 method–solid comparisons derive from 36 independent curves and 352 unique energies: the AE curves for BN and C are shared by both mixed protocols. Third-order Birch–Murnaghan fits give the equilibrium lattice constant and elastic response under isotropic compression.

The comparison uses the ten solids for which consistent EOS data were available in every representation. LiF and LiH are not included. The selection is identical across protocols and was not based on the magnitude of the reference error.

S8.2 Computational details and fit stability

All selected points use 800/60 Ry cutoffs, 150 radial points, the 770-point Lebedev rule, and the small extended one-centre basis. All are neutral, spin-restricted calculations with UZH 2026.2 basis and potential definitions.¹

GAPW-AE uses TZVPP-MOLOPT-PBE-ae orbital bases and explicit core electrons for every element. GAPW-XC/GTH uses TZV2P-MOLOPT-PBE-GTH bases and the corresponding element-specific GTH-PBE potentials. In the mixed protocols, Li, B, C, N, and O retain the AE TZVPP basis, while Mg, Al, Si, P, S, and Cl use the GTH TZV2P basis. The AE one-centre reconstruction is retained in both mixed protocols. Their direct/one-centre distinction concerns the GTH fields. Every reported energy includes D3(BJ). Diagonalization and OT use residual thresholds of 5×10^{-6} and 5×10^{-7} , respectively. Exact inputs specify the remaining dispersion and integration conventions.

The actual Gamma-centred meshes are listed in Table S25. The mesh is identical across representations at each volume but can change with volume within an EOS. Consequently, a smooth fit is not by itself evidence of k-point convergence. The selected full/reduced-mesh checks in the preceding section constrain symmetry reduction for their tested cases, not all possible geometries and restarts.

Every full-window minimum is bracketed. The largest fit RMS residual is 0.674 meV per atom. Omitting one endpoint changes the fitted lattice constant by at most 0.002444 Å (mixed-direct MgS, smallest-volume point omitted) and the bulk modulus by at most 9.997 GPa (GAPW-XC AlN, largest-volume point omitted). The smallest-volume point is essential to bracketing the LiCl minimum. These tests assess fit-window sensitivity. They do not measure basis or quadrature errors and are not transferable error bars for other curves.

S8.3 Basis sensitivity of structural observables

The ten independent AE curves contain 98 accepted TZVPP energy points. No matched QZVPP EOS series is included for these solids. Structural basis convergence must be assessed over volume, separately from SCF convergence and fit stability. For a basis change,

$$\delta E(V) = E_{\text{QZVPP}}(V) - E_{\text{TZVPP}}(V), \quad (\text{S6})$$

a constant δE changes only the energy origin. Its volume dependence can instead alter the equilibrium volume, curvature, and higher derivatives that determine the elastic response. An accurate fit to a finite-basis curve therefore does not establish basis-converged lattice constants or bulk moduli.

The paired Urea test in Sec. S6.3 identifies a substantial basis contribution to a crystal–molecule energy difference. Its magnitude cannot be transferred directly to the present structural observables, which contain no gas-phase reference. The ten-solid lattice contraction and elastic parameters are consequently reported as finite-basis predictions. This qualification also applies to the AE components of the mixed protocols. The shared BN and C curves are counted once and do not constitute independent basis checks.

S8.4 Representation and material trends

All native lattice constants are below the static-lattice experimental reference. LiCl contracts most, by approximately 0.154–0.164 Å, while C and BN contract by about 0.015–0.026 Å. MgS is also strongly contracted. The common sign spans diamond, zincblende, and rocksalt structures rather than a single bonding class.

GAPW-XC and AE differ on average by 0.01893 Å in a_0 and 10.55 GPa in B_0 , but this comparison changes basis and core treatment as well as density representation. At fixed mixed-core Hamiltonian, the one-centre correction changes a_0 by 0.000135 Å on average and at most 0.000542 Å (AlP). The corresponding changes in B_0 average 1.11 GPa and reach 5.75 GPa (MgO). Excluding the two shared AE curves gives mean changes of 0.000169 Å and 1.39 GPa over eight independent pairs. The main contraction trend is therefore insensitive to this representation choice.

Curvature errors are less uniform. GAPW-XC underestimates the Si bulk modulus, 92.76 versus 100.3 GPa, despite a contracted lattice, whereas AE gives 100.56 GPa. For C, GAPW-XC is closer to the reference modulus than AE. A shorter equilibrium lattice does not necessarily imply a larger bulk modulus, and AE does not improve every property of every material. Neither a_0 nor B_0 determines a cohesive energy without atomic references.

S8.5 Matched literature comparisons

All methods are compared against the same primary static-lattice experimental values from Goldzak *et al.*¹³ Their Hartree–Fock (HF), second-order Møller–Plesset perturbation theory (MP2), spin-component-scaled MP2 (SCS-MP2), and scaled opposite-spin MP2 (SOS-MP2) results cover all ten selected solids. Zhang *et al.*¹⁴ supply local-density approximation (LDA), PBE, PBEsol, M06-L, SCAN, and HSE06 values for nine, excluding AlN. We use their uncorrected static electronic values, not predictions with zero-point corrections. Their SCAN calculations were non-self-consistent on PBE densities. Schimka *et al.*¹⁵ provide PBE, PBEsol, HSE06, and HSEsol lattice constants for nine, excluding MgS. No missing bulk modulus is inferred. Self-consistent Tao–Mo results¹⁶ cover eight, excluding AlN and BN, and are rescored against the same Goldzak reference.

Our periodic GFN manuscript¹⁷ supplies GFN1-xTB and GFN2-xTB lattice constants for nine shared solids, but no bulk moduli in the curated comparison table. Its ten-material set excludes MgO and LiH, whereas the present set excludes LiF and LiH. The two ten-material selections must therefore not be treated as identical.

Table S16: Selected a_0 values (Å). Experimental references follow Ref. 13. BN and C reuse the AE curve for HD and HOC. These entries are not independent calculations.

Solid	Experiment	GX	AE	HD	HOC
AlN	4.3680	4.2811	4.3025	4.2747	4.2746
AlP	5.4480	5.3622	5.3781	5.3616	5.3621
BN	3.5930	3.5777	3.5675	3.5675	3.5675
BP	4.5250	4.4439	4.4696	4.4527	4.4527
C	3.5530	3.5345	3.5292	3.5292	3.5292
LiCl	5.0720	4.9081	4.9155	4.9178	4.9179
MgO	4.1890	4.1059	4.0935	4.0923	4.0919
MgS	5.1880	5.0689	5.0536	5.0700	5.0697
Si	5.4210	5.3386	5.3740	5.3383	5.3383
SiC	4.3470	4.2517	4.2920	4.2556	4.2556

Table S17: Selected B_0 values (GPa). Experimental references follow Ref. 13. BN and C reuse the AE curve for HD and HOC. These entries are not independent calculations.

Solid	Experiment	GX	AE	HD	HOC
AlN	206.00	239.07	225.89	240.42	240.40
AlP	87.00	94.34	101.35	96.36	93.12
BN	388.50	423.39	420.87	420.87	420.87
BP	176.50	187.93	183.64	185.86	185.86
C	453.30	457.33	491.35	491.35	491.35
LiCl	37.30	47.98	46.37	47.13	47.16
MgO	173.00	188.28	198.71	213.13	207.37
MgS	81.00	87.90	90.27	84.17	86.22
Si	100.30	92.76	100.56	94.13	94.13
SiC	228.90	263.89	241.68	263.73	263.71

SCS-MP2 and SOS-MP2 give matched ten-solid lattice MAEs of 0.0125 and 0.0149 Å, and bulk-modulus MAEs of 4.10 and 3.88 GPa. The native ranges are 0.0728–0.0844 Å and 16.61–21.77 GPa. Established semilocal and screened-hybrid methods also have smaller lattice errors on the corresponding intersections. These results identify a limitation of the present structural predictions. The comparisons do not uniquely separate functional and numerical contributions.

Alternative experimental moduli listed by Goldzak *et al.* for BN, BP, and MgO are 410.2, 168.0, and 169.8 GPa, instead of the primary 388.5, 176.5, and 173.0 GPa. Simultaneously replacing all three lowers each native ten-solid modulus MAE by 1.00 GPa and leaves the qualitative conclusion intact. Broader 44-solid statistics, such as those of Tran *et al.*,¹⁸ are not relabelled as matched ten-solid errors.

Table S18: EOS fit diagnostics, part 1. All full-window minima are bracketed. RMS residuals are in meV per atom.

Method	Solid	N_V	B'_0	RMS
AE	AlN	9	3.707	0.0418
AE	AlP	9	4.179	0.0880
AE	BN	10	3.772	0.1250
AE	BP	10	3.701	0.0311
AE	C	10	3.756	0.0542
AE	LiCl	10	4.029	0.0205
AE	MgO	10	4.092	0.0190
AE	MgS	10	3.682	0.0117
AE	Si	10	4.544	0.3496
AE	SiC	10	3.854	0.0449
GX	AlN	9	2.802	0.6739
GX	AlP	9	3.959	0.1431
GX	BN	10	3.713	0.1306
GX	BP	10	4.081	0.0723
GX	C	10	3.611	0.1734
GX	LiCl	10	4.027	0.0435
GX	MgO	10	3.864	0.2921
GX	MgS	10	4.349	0.1597

Table S19: EOS fit diagnostics, part 2. All full-window minima are bracketed. RMS residuals are in meV per atom.

Method	Solid	N_V	B'_0	RMS
GX	Si	10	5.096	0.5702
GX	SiC	10	4.107	0.1309
HD	AlN	9	3.339	0.3028
HD	AlP	9	4.778	0.3362
HD	BP	10	3.918	0.0532
HD	LiCl	10	3.905	0.0330
HD	MgO	10	4.208	0.3562
HD	MgS	10	3.167	0.2585
HD	Si	10	5.257	0.3634
HD	SiC	10	4.168	0.0561
HOC	AlN	9	3.338	0.3027
HOC	AlP	9	3.787	0.2878
HOC	BP	10	3.918	0.0532
HOC	LiCl	10	3.915	0.0328
HOC	MgO	10	4.181	0.0393
HOC	MgS	10	3.865	0.0993
HOC	Si	10	5.257	0.3634
HOC	SiC	10	4.167	0.0557

Table S20: Matched comparison with Ref. 13: ten-solid set. Every row uses the identical intersection and the Goldzak static-lattice experimental reference. $\text{MAE}(a_0)$ is in Å and $\text{MAE}(B_0)$ in GPa. A dash denotes unavailable data, not zero error. Native rows retain the numerical qualifications discussed in the text.

Method	N	$\text{MAE}(a_0)$	$\text{MAE}(B_0)$
GX	10	0.0831	16.61
AE	10	0.0728	16.89
HD	10	0.0844	21.77
HOC	10	0.0844	21.07
HF	10	0.0563	15.23
MP2	10	0.0201	5.71
SCS-MP2	10	0.0125	4.10
SOS-MP2	10	0.0149	3.88

Table S21: Matched comparison with Ref. 14: nine solids, excluding AlN. Every row uses the identical intersection and the Goldzak static-lattice experimental reference. $\text{MAE}(a_0)$ is in Å and $\text{MAE}(B_0)$ in GPa. A dash denotes unavailable data, not zero error. Native rows retain the numerical qualifications discussed in the text.

Method	N	$\text{MAE}(a_0)$	$\text{MAE}(B_0)$
GX	9	0.0827	14.78
AE	9	0.0737	16.56
HD	9	0.0834	20.36
HOC	9	0.0835	19.59
LDA	9	0.0324	4.66
PBE	9	0.0451	12.96
PBEsol	9	0.0113	4.72
M06-L	9	0.0128	3.46
SCAN	9	0.0123	3.08
HSE06	9	0.0170	5.86

Table S22: Matched comparison with Ref. 15: nine solids, excluding MgS. Every row uses the identical intersection and the Goldzak static-lattice experimental reference. $\text{MAE}(a_0)$ is in Å and $\text{MAE}(B_0)$ in GPa. A dash denotes unavailable data, not zero error. Native rows retain the numerical qualifications discussed in the text.

Method	N	$\text{MAE}(a_0)$	$\text{MAE}(B_0)$
GX	9	0.0791	–
AE	9	0.0660	–
HD	9	0.0807	–
HOC	9	0.0807	–
PBE	9	0.0438	–
PBEsol	9	0.0136	–
HSE06	9	0.0132	–
HSEsol	9	0.0117	–

Table S23: Matched comparison with Ref. 16: eight solids, excluding AlN and BN. Every row uses the identical intersection and the Goldzak static-lattice experimental reference. $\text{MAE}(a_0)$ is in Å and $\text{MAE}(B_0)$ in GPa. A dash denotes unavailable data, not zero error. Native rows retain the numerical qualifications discussed in the text.

Method	N	$\text{MAE}(a_0)$	$\text{MAE}(B_0)$
GX	8	0.0911	12.27
AE	8	0.0797	14.58
HD	8	0.0907	18.86
HOC	8	0.0907	18.00
Tao-Mo	8	0.0194	4.26

Table S24: Matched comparison with Ref. 17: nine solids, excluding MgO. Every row uses the identical intersection and the Goldzak static-lattice experimental reference. $\text{MAE}(a_0)$ is in Å and $\text{MAE}(B_0)$ in GPa. A dash denotes unavailable data, not zero error. Native rows retain the numerical qualifications discussed in the text.

Method	N	$\text{MAE}(a_0)$	$\text{MAE}(B_0)$
GX	9	0.0831	–
AE	9	0.0703	–
HD	9	0.0831	–
HOC	9	0.0830	–
GFN1-xTB	9	0.1360	–
GFN2-xTB	9	0.0659	–

Table S25: Actual Gamma-centered k meshes by volume index. Indices 01–10 refer to the original evenly spaced volume ratios 0.90–1.10. Each selected volume uses the same mesh in all representations, but the mesh can change along an EOS.

Solid	Mesh and volume indices
AlN	5 ³ : 01,02,03,04,05,06,07,08,09
AlP	4 ³ : 01,02,03,04,05,06,07,08,09
BN	6 ³ : 02,03,04,05,06,07,08,09,10; 7 ³ : 01
BP	5 ³ : 01,02,03,04,05,06,07,08,09,10
C	6 ³ : 04,05,06,07,08,09,10; 7 ³ : 01,02,03
LiCl	5 ³ : 01,02,03,04,05,06,07,08,09,10
MgO	5 ³ : 06,07,08,09,10; 6 ³ : 01,02,03,04,05
MgS	4 ³ : 07,08,09,10; 5 ³ : 01,02,03,04,05,06
Si	4 ³ : 02,03,04,05,06,07,08,09,10; 5 ³ : 01
SiC	5 ³ : 01,02,03,04,05,06,07,08,09,10

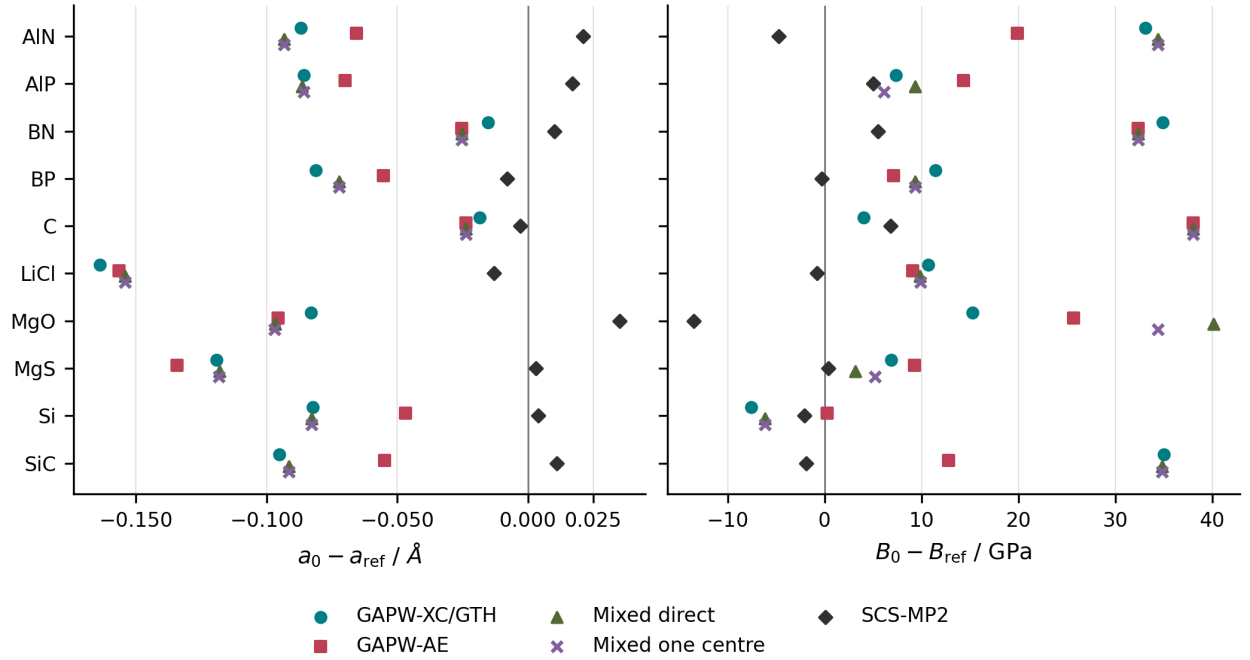


Figure S1: Signed lattice-constant and bulk-modulus errors against the common Goldzak static-lattice reference. SCS-MP2 provides matched correlated-method context. Systematic native lattice contraction persists across representations. Its sign alone does not identify the source of the discrepancy.

S9 Pressure Derivatives and Compressibilities

The third-order Birch–Murnaghan fit supplies V_0 , B_0 , and $B'_0 = (\partial B/\partial P)_{P=0}$ in addition to E_0 . Its energy expression is

$$E(V) = E_0 + \frac{9V_0B_0}{16} [B'_0(x-1)^3 + (6-4x)(x-1)^2], \quad x = (V_0/V)^{2/3}. \quad (\text{S7})$$

The zero-pressure compressibility is $\kappa_0 = 1/B_0$. These quantities require no new SCF calculation. B'_0 depends on the third derivative of the fitted energy and is more sensitive to the fit interval and numerical noise than the minimum location. We therefore supplement the full-window values with endpoint-omission sensitivity rather than assigning artificial high precision from the least-squares fit.

Table S26: Additional predictions from the existing EOS data. κ_0 is in 10^{-3} GPa^{-1} . V_0 is the eight-atom cell volume in $\text{\AA}^3$. The endpoint column is the largest absolute change of B'_0 after removing either endpoint. A dagger means at least one omitted-window minimum is no longer bracketed.

Method	Solid	V_0	B_0/GPa	B'_0	$10^3\kappa_0$	max. $ \Delta B'_0 $
AE	AlN	79.6434	225.89	3.707	4.427	0.061
AE	AlP	155.5561	101.35	4.179	9.867	0.292
AE	BN	45.4045	420.87	3.772	2.376	0.106
AE	BP	89.2889	183.64	3.701	5.445	0.050
AE	C	43.9583	491.35	3.756	2.035	0.092
AE	LiCl	118.7693	46.37	4.029	21.568	0.073 [†]
AE	MgO	68.5923	198.71	4.092	5.032	0.049
AE	MgS	129.0645	90.27	3.682	11.077	0.031
AE	Si	155.2040	100.56	4.544	9.944	0.921
AE	SiC	79.0664	241.68	3.854	4.138	0.032
GX	AlN	78.4652	239.07	2.802	4.183	1.895
GX	AlP	154.1793	94.34	3.959	10.600	0.466
GX	BN	45.7944	423.39	3.713	2.362	0.129
GX	BP	87.7590	187.93	4.081	5.321	0.038
GX	C	44.1546	457.33	3.611	2.187	0.350
GX	LiCl	118.2330	47.98	4.027	20.844	0.230 [†]
GX	MgO	69.2214	188.28	3.864	5.311	0.625
GX	MgS	130.2384	87.90	4.349	11.377	0.264
GX	Si	152.1509	92.76	5.096	10.780	0.810
GX	SiC	76.8604	263.89	4.107	3.789	0.230
HD	AlN	78.1094	240.42	3.339	4.159	0.634
HD	AlP	154.1285	96.36	4.778	10.378	0.750
HD	BN	45.4045	420.87	3.772	2.376	0.106
HD	BP	88.2826	185.86	3.918	5.380	0.091
HD	C	43.9583	491.35	3.756	2.035	0.092
HD	LiCl	118.9380	47.13	3.905	21.217	0.212 [†]
HD	MgO	68.5343	213.13	4.208	4.692	0.337
HD	MgS	130.3228	84.17	3.167	11.880	0.631
HD	Si	152.1244	94.13	5.257	10.624	1.074
HD	SiC	77.0688	263.73	4.168	3.792	0.099
HOC	AlN	78.1082	240.40	3.338	4.160	0.634
HOC	AlP	154.1753	93.12	3.787	10.739	0.766
HOC	BN	45.4045	420.87	3.772	2.376	0.106
HOC	BP	88.2820	185.86	3.918	5.381	0.091
HOC	C	43.9583	491.35	3.756	2.035	0.092
HOC	LiCl	118.9405	47.16	3.915	21.203	0.240 [†]
HOC	MgO	68.5140	207.37	4.181	4.822	0.070
HOC	MgS	130.2985	86.22	3.865	11.598	0.224
HOC	Si	152.1238	94.13	5.257	10.624	1.074
HOC	SiC	77.0679	263.71	4.167	3.792	0.099

The full-window B'_0 values span 2.80–5.26. The largest endpoint-induced change, 1.90, occurs for GAPW-XC AlN, whose full-window value is 2.80. This substantial sensitivity cautions against assigning physical significance to small differences between fitted pressure derivatives. The accompanying machine-readable analysis retains both endpoint tests and identifies cases in which an omitted endpoint no longer brackets the minimum.

The matched primary literature tables provide a_0 and B_0 but no consistently curated common B'_0 reference set. We report B'_0 as a derived model prediction with a fit-window diagnostic, not an accuracy ranking against an invented reference. Compressibility comparisons follow directly from the same matched B_0 values. They contain no independent evidence beyond that observable. The three molecular crystals have only paired single-point energies, so no bulk modulus, pressure derivative, elastic tensor, or equilibrium volume is inferred for them.

S10 Reproducibility and Numerical Limits

The molecular, molecular-crystal, ice, and solid subsets use distinct documented protocols and are analysed separately. Each reported energy is linked to its executed input, model and executable versions, SCF convergence, and normal termination record. Replicate calculations count once. The archive distinguishes production energies from the independent numerical controls and preserves the input and source hashes needed to reproduce the reported quantities. The molecular QZVPP ethane outlier remains explicitly identified as unresolved despite normal SCF termination. The preliminary ice subset is not presented as a completed DMC-ICE13 benchmark.

The ESI reports available convergence and sensitivity data rather than claiming a universal bound on basis, integration, or k-point error. In particular, the present work does not provide complete-basis extrapolation, BSSE correction for the molecular crystals, converged phonons, finite-temperature thermodynamics, or cohesive energies for the ten solids. Such quantities would require additional calculations and reference choices. The reported lattice-energy and structural trends remain useful, provided those distinctions are maintained.

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